Thermally Reversible Beta-Sheet-like Ordering Controls Gelation in Elastin-like Polypeptides

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Abstract

Amyloid-like β-sheet packing spans a wide range of stabilities, from the kinetically persistent cores of pathological fibrils to contacts that come apart again, and where a sequence falls in that range decides whether it can serve as a switchable material. Here, temperature-dependent small- and wide-angle X-ray scattering of the sequence-defined elastin-like polypeptide $(VPAVG)_{30}$ resolves a transition between a dispersed low-temperature state and a hierarchical high-temperature assembly carrying β-sheet-like peaks at $Q = 1.38$ and 0.72 Å^{-1} that vanish on cooling. Separating what is slow

from what is permanent takes two measurements. On a continuous cooling ramp none of three chain lengths returns to its heating branch, so the hysteresis of a ramp is a rate effect and says nothing about reversibility. Given time the two outcomes diverge: $(VPAVG)_{30}$ recovers its low-temperature profile to within 9% through seven held cycles, while the ten-repeat construct stays turbid after two days at 4 °C. Chain length therefore decides whether the assembly is a physical junction or a terminal aggregate.

Introduction

Amyloid-like β-sheet ordering occupies an unusual position in protein science and polymer materials. Pathological amyloid fibrils formed by proteins and peptides that share no sequence homology are associated with Alzheimer's disease,[1] Parkinson's disease,[2] type II diabetes,[3] prion diseases,[4] and systemic light-chain amyloidosis.[5] Many amyloid fibrils are kinetically persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-zipper interfaces.[6] Yet amyloid-like order is not intrinsically pathological: functional amyloids store mammalian peptide hormones in pituitary secretory granules[7] and build the extracellular matrix of bacterial biofilms,[8] and related assemblies are increasingly used as mechanically robust biomaterials.[9] What differs across these systems is not whether β-sheet-like order forms but how long it lasts: hormone granules disassemble to release functional monomer, whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a wide range of stabilities rather than committing a sequence to one permanent structure,[10] and where a sequence falls in that range is a materials-design variable.

Reversible β-sheet-like contacts are attractive as candidate physical junctions because backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or weakly packed interfaces can remain dynamic or reversible.[10–13] Elastin-like polypeptides (ELPs) are well suited to test this possibility because their sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly.[14–18] Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or

hierarchical assemblies depending on sequence, concentration, and thermal history.[17–19] In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested phase separation.[20,21] Although these studies establish the arrested state from time-averaged scattering and rheology, how the assembly evolves toward arrest remains elusive.

Results and Discussion

Figure 1 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were measured at APS 12-ID-B by simultaneous SAXS/WAXS at 13.3 keV.[22–24] The "Measurement" aliquot had undergone more than 10 heating/cooling cycles; the "Reference" aliquot had stayed at 4 °C. At 10 °C the two overlap, and a Guinier fit to the Reference gives an apparent $R_g = 22.65 \pm 0.50$ Å (SI Appendix), a dispersed state with no mesoscale aggregates down to $Q = 0.0081$ Å^{-1}, which corresponds to structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(Q) \propto Q^{-3.2}$ response at low Q (slopes -3.17 and -3.15 over $0.012 < Q < 0.040$ Å^{-1}). Both also develop the same pair of high-Q features, at $Q = 1.38$ and 0.72 Å^{-1} in the Reference and 1.36 and 0.74 Å^{-1} in the Measurement. The cycled aliquot alone makes the point: it still develops both peaks at 30 °C and shows neither at 10 °C. A more direct test at 8-ID-I took a single aliquot through seven consecutive 6–34 °C cycles, heated at 1 °C min^{-1} (SI Appendix). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation with no systematic drift, and at 33.8 °C the correlation function stays close to $1 + \beta$ in every cycle, the signature of an arrested state. The sample therefore assembled and arrested on every heating and returned to the same low-temperature state on every cooling (SI Appendix). Video S1 independently shows recovery of optical transparency after cooling.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle scattering signifies density heterogeneity extending across the probed interval rather than a single resolved structural scale.[25] Related gel-forming ELPs exhibit disordered bicontinuous

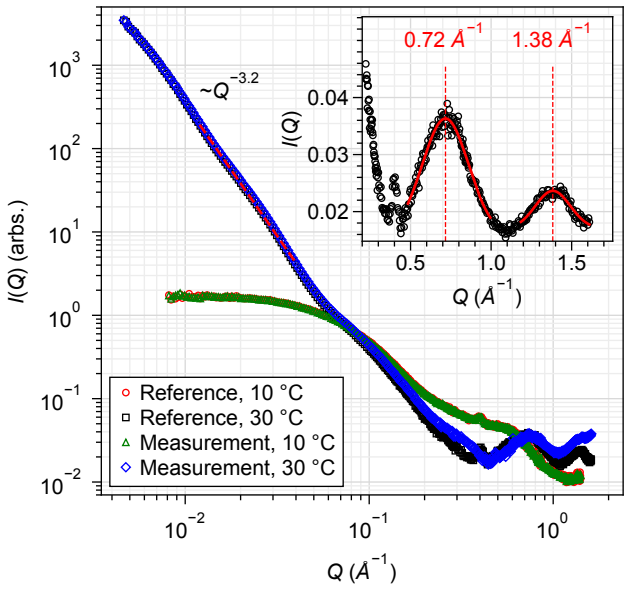


Figure 1: Temperature-dependent structure and reversibility of $(\mathrm{VPAVG})_{30}$. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50$ Å (SI Appendix). Heating produces an approximate $Q^{-3.2}$ low-Q response and peaks at $Q = 1.38$ and 0.72 Å^{-1}, corresponding to real-space spacings of approximately 4.5 and 8.8 Å, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < Q < 0.040$ Å^{-1}. Inset: the high-Q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70$ Å^{-1}, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

or fractal protein-rich networks produced by arrested phase separation.[20,21] The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross-β-like assemblies, whose reflections are reported at approximately 4.8 Å along the fibril direction and between 8 and 11 Å across it.[6,26] Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 Å^{-1} (correlation lengths $2\pi/\Delta Q$ of only about 25 and 19 Å), so the ordering extends over a few repeat spacings rather than forming extended cross-β crystallites. Their concurrent appearance with the low-Q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity. The disappearance of the peaks on cooling identifies thermally labile β-sheet-like associations that are plausible, rather than uniquely established, physical junctions.

The comparison of Figure 1 sets two separate 12-ID-B aliquots against each other. The test in Figure 2 is more direct: one aliquot at 8-ID-I, taken through seven consecutive cycles and compared with itself, so that any difference between cycles can only be irreversibility. The aliquot is a separate loading of the same stock, measured in the 2022-1 beamtime. Each cycle held the sample at 6 °C, ramped to 34 °C at 1.01 °C min^{-1} over 270 consecutive 2 s acquisitions, and returned to 6 °C, taking 5.8 h in total. No acquisitions were taken during cooling, so the cooling segments of Figure 2a are a projection from the fixed control protocol rather than a measurement; every solid segment is measured, once per acquisition, from the RTD of the stage zone holding the cell.

Three states are sampled in every cycle: 6 °C, and two windows near the top of each ramp at 31.9 °C and 33.8 °C. Averaged over $0.004 < Q < 0.008$ Å^{-1}, the seven cycles agree to a relative standard deviation of 9% at 6 °C and 5% at 33.8 °C, with no drift from the first cycle to the seventh. At 31.9 °C the spread is 33%, but that reflects where on the ramp each cycle happened to be sampled rather than a difference between cycles: the windows fall between 31.62 °C and 32.48 °C, which is on the steep part of the transition. Regressing the band-averaged $\ln I$ of each cycle on that cycle's mean window temperature gives $\mathrm{d}\ln I/\mathrm{d}T = 0.95\,^{\circ}\mathrm{C}^{-1}$ with $r = 0.974$, and the spread taken about that line falls to 7%.

The same regression at 33.8 °C, past the steep part, takes 5% to 1%. The cycles therefore repeat to better than 10% at every state once the sampling temperature is accounted for.

The correlation functions say what state the sample was in when each was taken (Figure 2c). At 33.8 °C the correlation function shows no resolvable decay anywhere in the accessible window, from 10^{-2} to 2 s, in any of the seven cycles: the sample was arrested on every heating. At 31.9 °C it decays throughout that same window, so the assembly is still mobile 1.9 °C lower. The dashed line in that panel is the measured correlation level of a static reference recorded in the same beamtime, read at the same Q bin. A static sample cannot relax, so its level is the instrumental contrast, and showing it as a curve rather than as a fitted number keeps the comparison between two measurements. The 33.8 °C correlation functions sit just below it and the 31.9 °C ones fall well away from it. A contrast value is needed to fit a correlation function, which is why a fitted series carries its own calibration; it is not needed to judge whether a sample has arrested, which is what this panel does, so no arrested fraction is quoted from it. No correlation function is shown at 6 °C, where the coherent signal at the lowest Q is too weak for any contrast to be resolved.

Whether the assembly comes apart again is a question about rate as much as about temperature, and the two have to be separated before either can be read. Three chain lengths, $(\mathrm{VPAVG})_{30}$ and its twenty- and ten-repeat counterparts, were taken from 20 °C up and immediately back down on one stage, each reaching a different ceiling because one protocol meets three different phase diagrams: 45.2, 41.4 and 33.1 °C for ten, twenty and thirty repeats (Figure 3). On the way up all three build low-Q intensity and all three lose the decay of g_2, so all three assemble and arrest.

None of them comes back down the way it went up. Averaged over $0.004 < Q < 0.008$ Å^{-1}, the cooling branch at 25 °C and below stands 8.4, 15.7 and 3.6 times the heating branch for ten, twenty and thirty repeats, while at the top of each ramp the two branches agree to within 2 to 19%. The hysteresis is therefore in the cold state alone, and it does not order with chain length. A ramp that spends about twenty-five minutes between the ceiling and

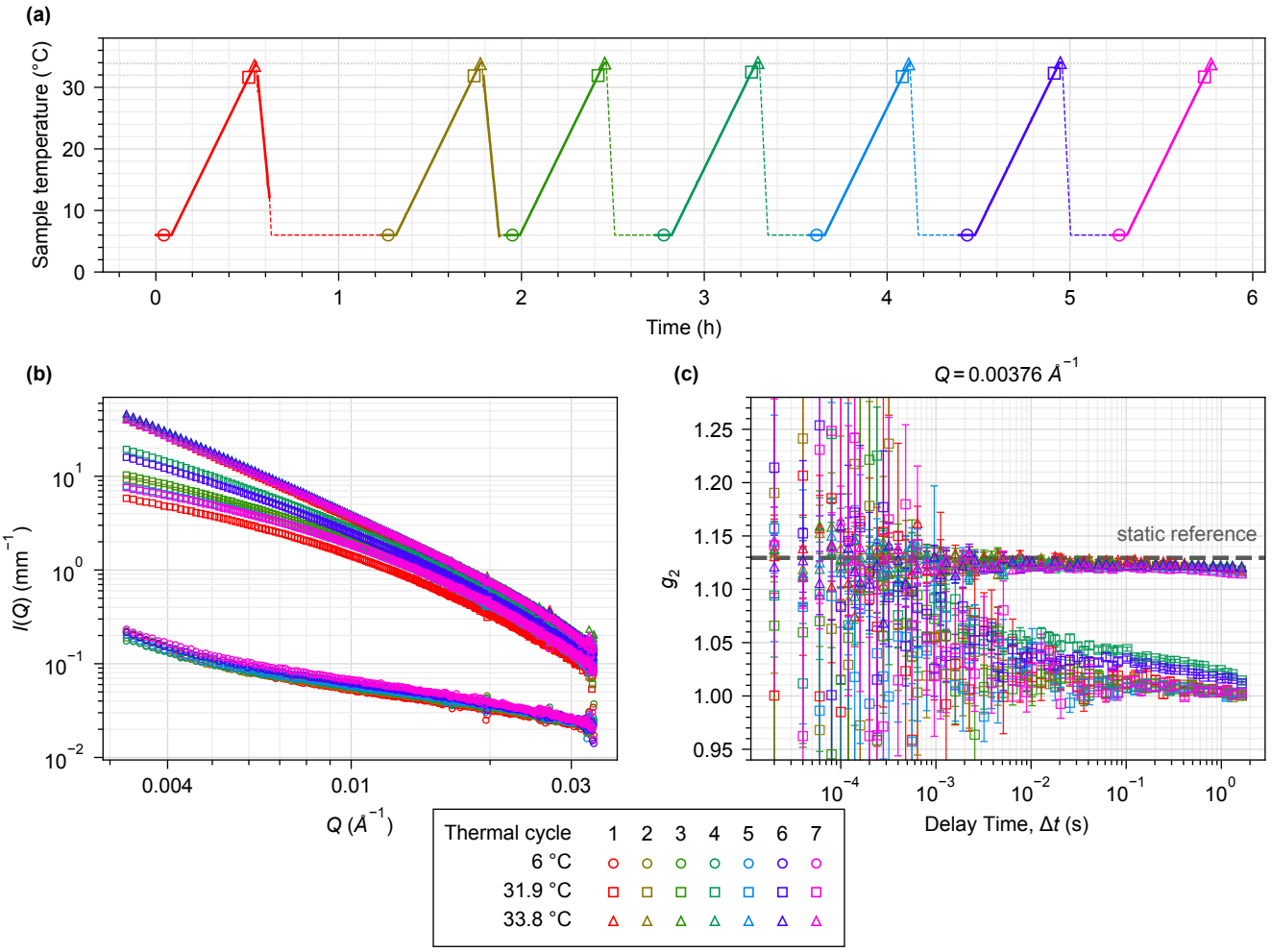


Figure 2: Thermal-cycling repeatability of the SA-XPCS sample: one aliquot through seven consecutive cycles. (a) Sample temperature over the 5.8 h sequence. Solid curves are measured: the stage RTD was read once per 2 s acquisition. No acquisitions were taken while the sample cooled, so no temperature was recorded over those intervals; the dashed curves there are not data but the profile projected from the control protocol, which was fixed at 10 °C min^{-1} cooling until 6 °C followed by a hold at 6 °C until the next ramp. Open symbols mark, at the time and temperature at which it was taken, every acquisition group that is plotted in (b) and (c), so each lower-panel curve can be traced to its point in the thermal history. (b) Buffer-subtracted SAXS on an absolute scale at 6 °C (circles, 32–41 surviving acquisitions averaged; see the text), 31.9 °C (squares, acquisitions 241–250) and 33.8 °C (triangles, acquisitions 261–270). Each group is placed on the absolute scale with the mean ion-chamber readings of the acquisitions it contains, and the averaged buffer (69 surviving of 89 acquisitions, D0077 and D0080) is subtracted with no empirical scale factor. (c) Correlation functions at the lowest Q bin for the two high-temperature states. The dashed line is the measured correlation level of a static reference recorded in the same beamtime and read at the same Q bin; a static sample cannot relax, so that level is the instrumental contrast. Both are measurements on one axis, so no calibration is imported and none is quoted. The 33.8 °C curves run flat just below the line across four decades of delay, which is arrest; the 31.9 °C curves decay away from it, which is not. Error bars are the measured pointwise g_2 uncertainties; the y axis is capped so that a few short-delay caps clip. One key serves all three panels: its columns are the seven thermal cycles, numbered along the top and colored on a single scale, and its rows are the three temperature states, labeled at the left and distinguished by marker shape.

20 °C simply does not give the assembly time to come apart, whatever the sequence. The hysteresis of a cooling ramp is a statement about rate, and on its own it says nothing about whether the transition is reversible.

What the sequence decides shows up once the cold state is given time, and then the three part company. Held at 6 °C between cycles, $(\mathrm{VPAVG})_{30}$ returns to the same 6 °C profile to within a 9% relative standard deviation through seven heating and cooling cycles, with no drift from the first to the seventh (Figure 2), and two aliquots of the same stock, one cycled more than ten times and one never heated, overlap at 10 °C (Figure 1). The ten-repeat construct does not recover at all: a cell loaded with it and left in a refrigerator at 4 °C for more than forty-eight hours after a single ramp is still visibly opaque, against the sixty seconds in which $(\mathrm{VPAVG})_{30}$ clears (Video S1). Two days is five orders of magnitude longer than the sixty seconds $(\mathrm{VPAVG})_{30}$ needs and three orders longer than the cooling ramp, so the difference between the two constructs is not a slower rate but a different outcome.

Taken together the two observations separate cleanly. Ramp hysteresis is universal and kinetic: every chain length shows it, it lives only in the cold state, and it reflects how fast the ramp was run. Reversibility is sequence-encoded: given time, thirty repeats come apart and ten do not. It is the second that makes the β-sheet-like contacts of $(\mathrm{VPAVG})_{30}$ a candidate physical junction rather than a terminal aggregate, and the first that explains why a single cooling ramp cannot be used to test for it.

Materials and Methods

To be assembled from the pre-split Letter (tag `pre-split` in PubCode-2021-IDPP): the 12-ID-B paragraph, the 8-ID-I cell and beam description, and the robotic cell-assembly method.

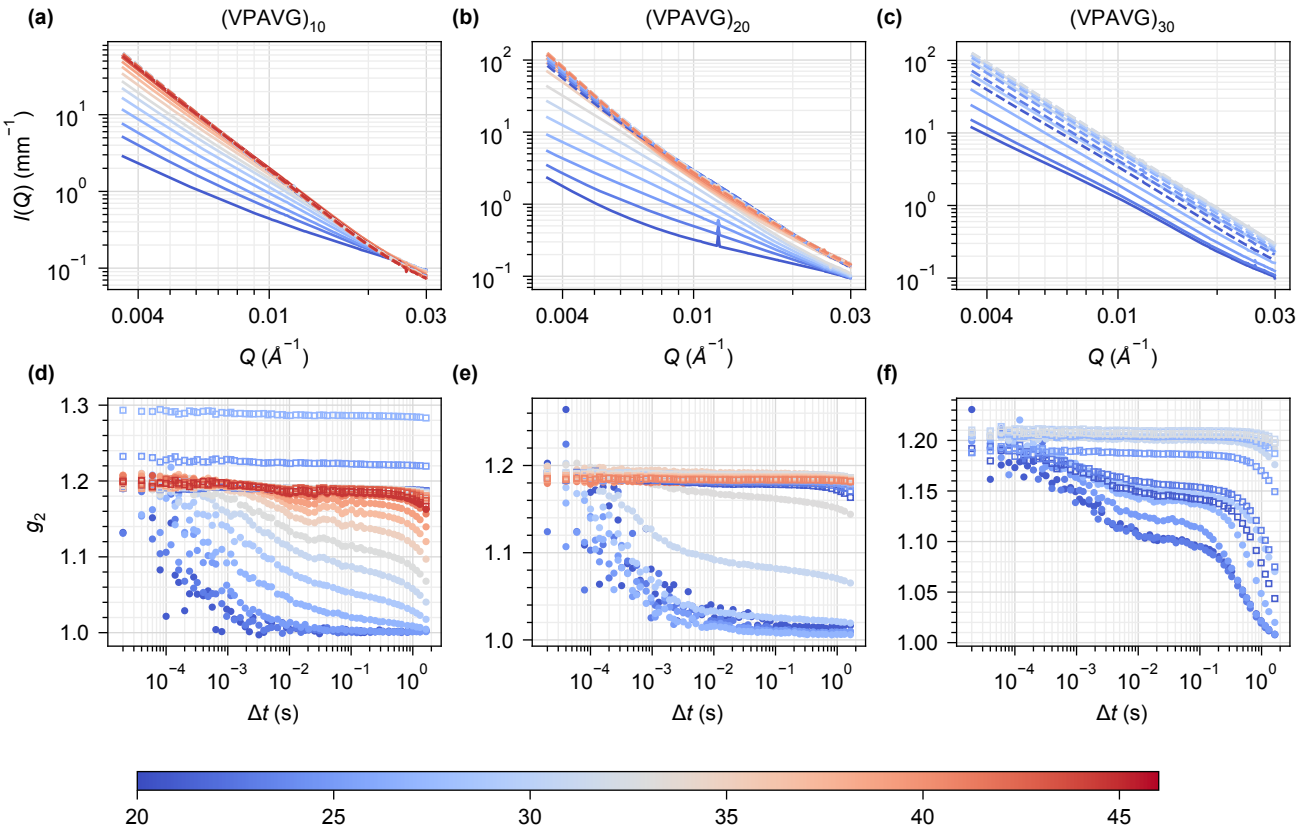


Figure 3: Three chain lengths through one thermal ramp, heating and cooling. Columns are $(\mathrm{VPAVG})_{30}$ with ten, twenty and thirty repeats. (a)–(c) background-subtracted $I(Q)$ on the absolute scale; (d)–(f) the intensity autocorrelation function at the lowest measured Q, the largest length scale. Solid lines and filled markers are the heating ramp, dashed lines and open markers the cooling ramp that followed it immediately. Colour is temperature on one scale shared by every panel, so a colour means the same temperature throughout; the ceiling differs between samples because one protocol was applied to three materials. Each curve averages the acquisitions falling in a 2 °C window, through the same two outlier cuts every other group in this work receives. The cooling branch lies above the heating branch at low temperature in all three samples, by 8.4, 15.7 and 3.6 times over $0.004 < Q < 0.008$ Å^{-1} below 25 °C; the branches meet again at the top of each ramp. That hysteresis is a rate effect common to all three and is not what distinguishes them; the held measurements of Figure 2 and Video S1 are. The absolute scale carries a common factor of about 6% from the transmission, which is taken from a different beamtime because 2021-2 recorded no downstream ion chamber; it is the same factor for all three samples and moves no feature within a profile.

Supporting Information

Recombinant synthesis and purification; sample environments and the automated cell assembly; SAXS/WAXS calibration and merging; Guinier analysis; chain-length ramps; seven-cycle thermal repeatability; irreversibility of the ten-repeat construct; videos of the reversible transition and of the robotic assembly (PDF, MP4).

Data and Code Availability

Reduced data and analysis code are at `https://github.com/qzhang234/PubCode-2021-IDPP-Reversibil`

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